Workshop A

MF2007

David Demir, Hsiu-Fang Chien, Johan Kindlundh
Group 11

February 3, 2026

Table of Contents

1 Parameter Identification **1**
1.1 Linear Friction Model 1
1.2 Nonlinear Friction Model 6

2 Controlling the Motor **11**
2.1 Velocity Control 11
2.2 Position Control 22

List of Acronyms and Abbreviations

DBE Design by Emulation.

DDD Discrete Digital Design.

DoF Degrees of Freedom.

FOH First Order Hold.

1 Parameter Identification

1.1 Linear Friction Model

This subsection derives and validates a linear model of the motor by estimating the parameters: linear friction d_m and load inertia J_l from experimental data. The motor is modelled as a linear system by neglecting static friction and terminal inductance, providing a simplified representation of the motor's behavior for Level 1 parameter identification.

1.1.1 Derivation of transfer function for position

From Kirchhoff's voltage law, the electric dynamics is described as Equation 1.1:

$$U - Ri - K_{emf}\dot{\varphi} = 0, \tag{1.1}$$

In the Laplace domain with an assumption of zero initial conditions, it gives Equation 1.2:

$$U(s) = RI(s) + K_{emf}s\Phi(s). \tag{1.2}$$

From Newton's second law, the mechanical dynamics is described as Equation 1.3

$$J_{tot}\ddot{\varphi} = K_m i - d_m\dot{\varphi}. \tag{1.3}$$

In the Laplace domain with an assumption of zero initial conditions, that is Equation 1.4:

$$J_{tot}s^2\Phi(s) = K_m I(s) - d_m s\Phi(s). \tag{1.4}$$

By isolating $I(s)$ from Equation 1.2 and substituting it into Equation 1.4, the transfer function from input voltage $U(s)$ to motor position $\Phi(s)$ is as Equation 1.5:

$$G_\varphi(s) = \frac{\Phi(s)}{U(s)} = \frac{\frac{K_m}{R}}{J_{tot}s^2 + \left(\frac{K_m K_{emf}}{R} + d_m\right)s} \tag{1.5}$$

This transfer function relates the input voltage to the motor's angular position, assuming linear dynamics.

1.1.2 Derivation of transfer function for velocity

Replacing the notation for the derivative of angular position $\dot{\varphi}$ with angular velocity ω from Equation 1.1 and transforming it into Laplace domain assuming zero initial conditions, gives Equation 1.6:

$$U(s) = RI(s) + K_{emf}\Omega(s). \tag{1.6}$$

Similarly, the replacement of ω and Laplace transform to Equation 1.3 gives Equation 1.7:

$$J_{tot}s\Omega(s) = K_m I(s) - d_m\Omega(s). \tag{1.7}$$

By isolating $I(s)$ from Equation 1.6 and substituting it into Equation 1.7, the transfer function from input voltage $U(s)$ to motor's angular velocity $\Omega(s)$ is as Equation 1.8:

$$G_\omega(s) = \frac{\Omega(s)}{U(s)} = \frac{\frac{K_m}{R}}{J_{tot}s + \left(\frac{K_m K_{emf}}{R} + d_m\right)} \tag{1.8}$$

1.1.3 Linear parameter identification

The DC gain of a transfer function can be determined using the final value theorem, defined as:

$$\text{Final Value} = \lim_{s\to 0} sG(s)\frac{1}{s} = \lim_{s\to 0} G(s). \tag{1.9}$$

Applying the final value theorem to the transfer functions in Equation 1.5 and Equation 1.8, the DC gain of position K_φ is obtained in Equation 1.10:

$$K_\varphi = \lim_{s\to 0} G_\phi(s) = \lim_{s\to 0} \frac{\Phi(s)}{U(s)} = \lim_{s\to 0} \frac{\frac{K_m}{R}}{J_{tot}s^2 + \left(\frac{K_m K_{emf}}{R} + d_m\right)s} = \infty. \tag{1.10}$$

The DC gain of velocity K_ω is as Equation 1.11:

$$K_\omega = \lim_{s\to 0} G_\omega(s) = \lim_{s\to 0} \frac{\frac{K_m}{R}}{J_{tot}s + \left(\frac{K_m K_{emf}}{R} + d_m\right)} = \frac{\frac{K_m}{R}}{\frac{K_m K_{emf}}{R} + d_m}. \tag{1.11}$$

The time constant τ for the velocity is derived from the denominator of Equation 1.8 and is given in Equation 1.12:

$$\tau = \frac{J_{tot}R}{K_m K_{emf} + Rd_m}. \tag{1.12}$$

To determine the motor parameters, the experimentally measured time constant $\bar{\tau}$ and DC gain $\bar{K}_\omega$ are used to derive the damping coefficient d_m and the load inertia J_l, which can then be tuned to accurately describe the motor's dynamics. With the known electrical parameters, Equations 1.11 and 1.12 are rearranged to isolate d_m and J_l as in Equations 1.13 and 1.14:

$$d_m = \frac{\frac{K_m}{\bar{K}_\omega} - K_m K_{emf}}{R} \tag{1.13}$$

$$J_l = \bar{\tau}\left(d_m + \frac{K_{emf}K_m}{R}\right) - J_m \tag{1.14}$$

The measured DC gain $\bar{K}_\omega$ is used to calculate the damping coefficient d_m with Equation 1.13. Since the time constant $\bar{\tau}$ depends on both d_m and J_l, the calculated d_m is then substituted into Equation 1.14 to determine J_l.

The nonlinear damping behaviour of the motor in the lab combined with the linear damping assumption in the model means that the model can only be fitted accurately to a single input amplitude. Since the goal is to design a robust controller based on the model, the parameters are chosen to represent the input range $[5, 20]$ volts. The amplitudes of 5, 12, and 20 volts were selected to cover the operational range of 0–24 volts while ensuring the PWM duty cycle remains within 10–90%. This range avoids issues such as insufficient torque at very low duty cycles or signal saturation at very high duty cycles, which could compromise the accuracy of the experimental data.

The time constant $\bar{\tau}$ and DC gain $\bar{K}_\omega$ are obtained from the experimental runs as shown in Figures 1–3. The parameters d_m and J_l are chosen to match these values as closely as possible. However, the experimental data reveals nonlinear dynamics that the linear model cannot fully capture. For instance, as shown in Table 1, the DC gain varies for different input amplitudes, suggesting that the motor does not behave as a purely linear system. Similarly, but less noticeable, discrepancies in the time constant also indicate underlying dynamics not captured by the model.

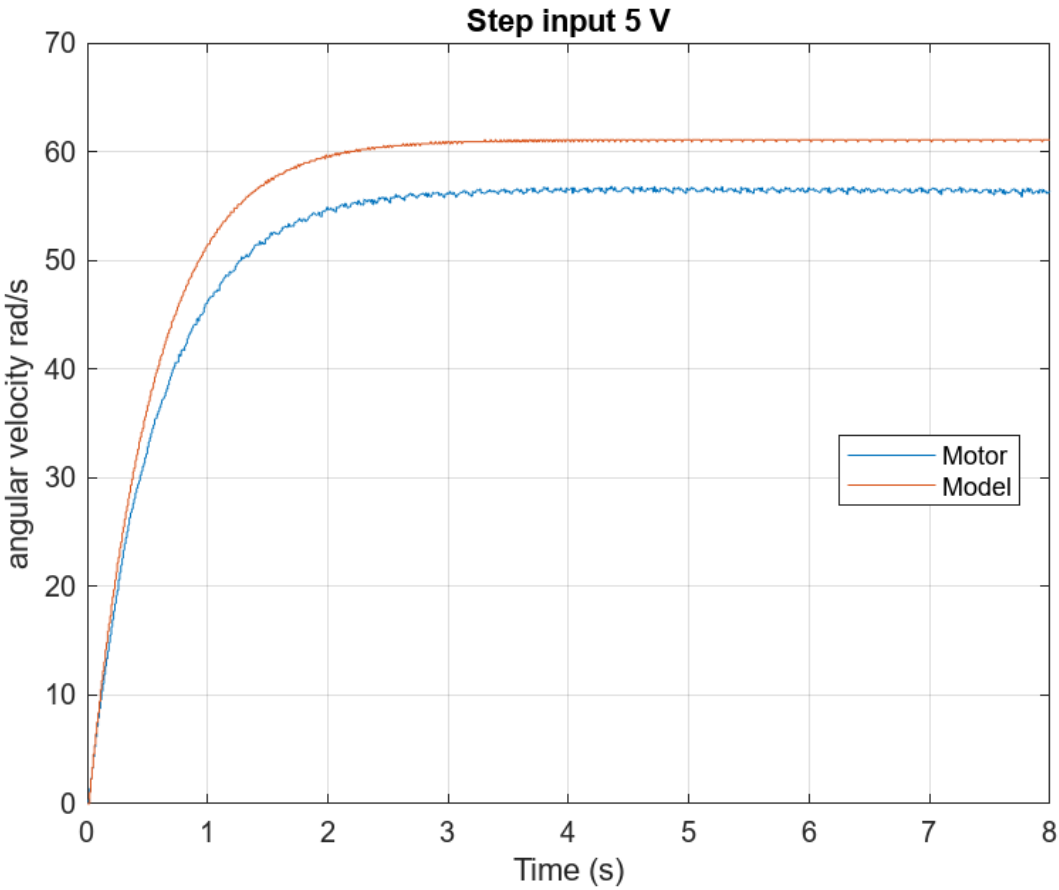

Figure 1: Comparison between motor and model with $u = 5$ V.

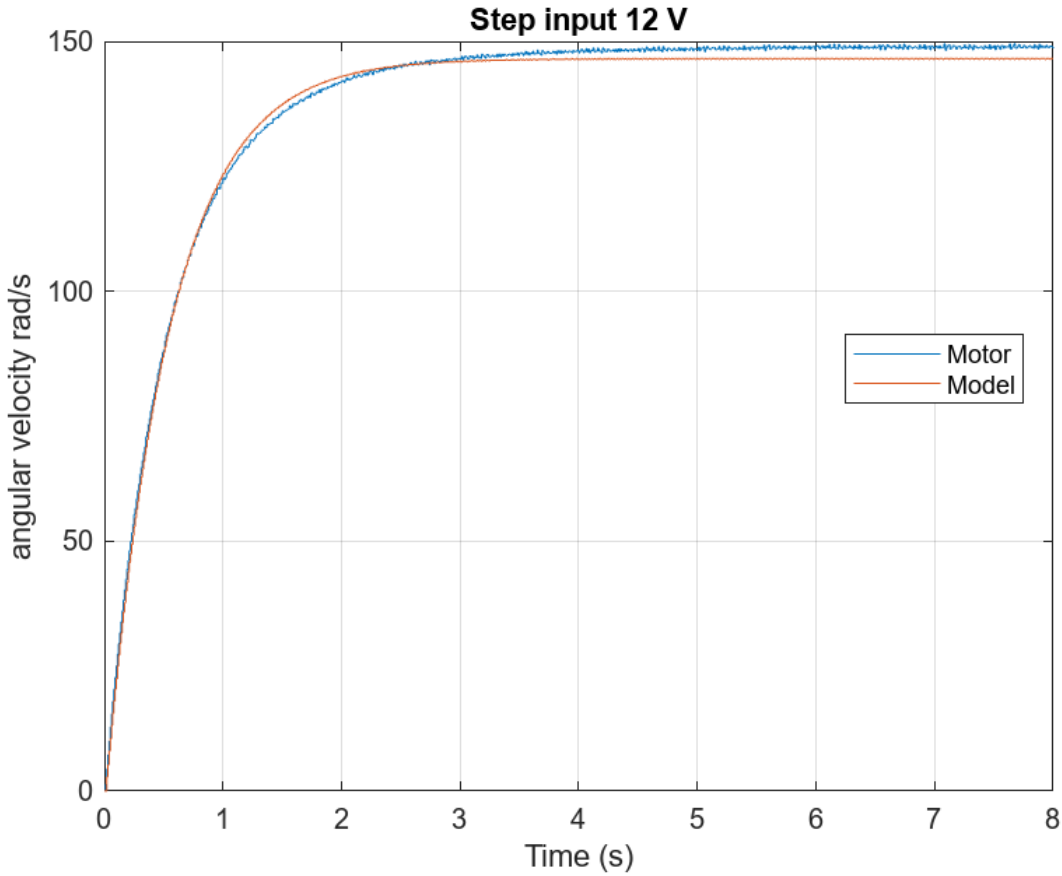

Figure 2: Comparison between motor and model with $u = 12$ V.

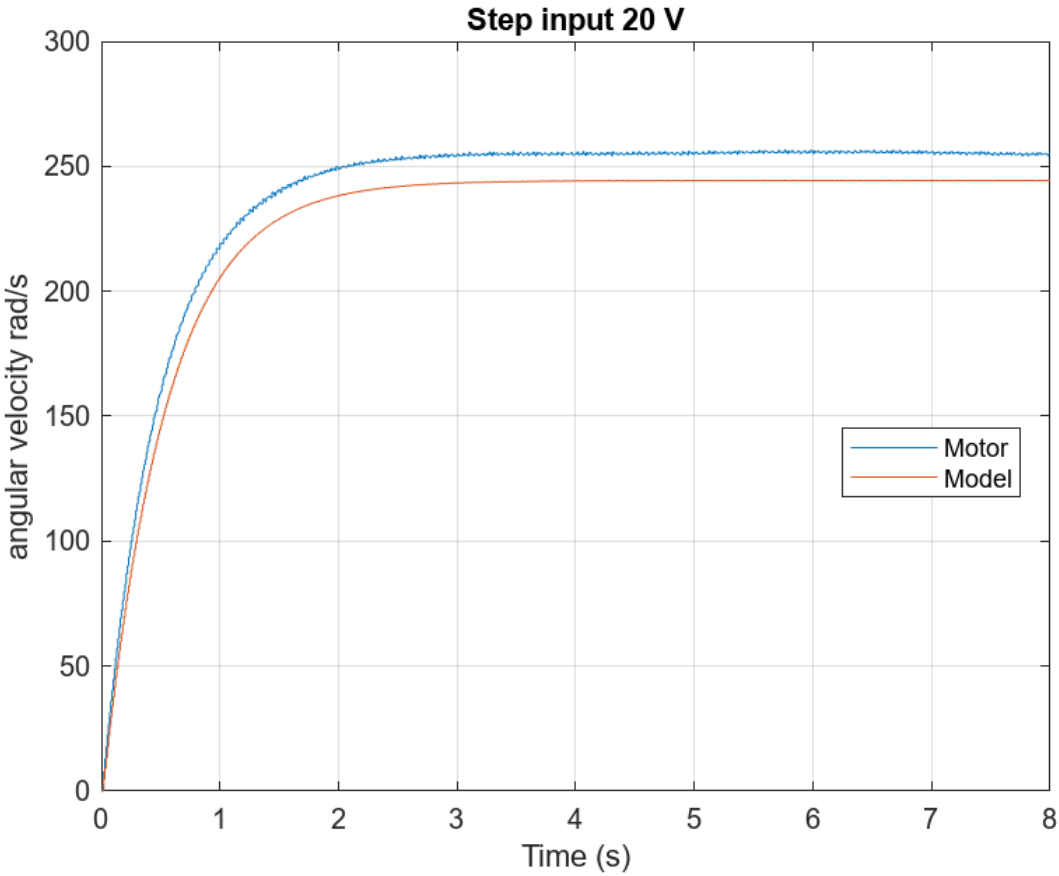

Figure 3: Comparison between motor and model with $u = 20$ V.

Table 1 displays the respective time constant and DC gain with different step inputs.

Table 1: Comparison of DC gains and time constants for motor and model.

Input (V)	DC Gain (Motor)	DC Gain (Model)	τ (Motor)	τ (Model)
5	11.3	12.2	0.58	0.55
12	12.3	12.2	0.55	0.55
20	11.8	12.2	0.51	0.55

From both the plots in Figures 1–3 and Table 1, it is evident that the linear model lacks the fidelity required to fully represent the real system's dynamics. Key factors missing from the model include static friction and nonlinear damping effects. The use of lumped parameters in a linear model provides only a simplified representation of the motor's behavior, which is insufficient for accurately predicting control performance based on this model.

The approach to fitting the model aimed to match all three step input amplitudes (5 V, 12 V, and 20 V) equally well. However, this resulted in a compromise, where the model

failed to accurately capture the motor's behavior for any specific input amplitude. The parameters were derived to approximate the system's dynamics within the selected voltage range as shown in Table 2.

Table 2: Identified parameters of linear model.

d_m (N m s rad^{-1})	J_l (kg m^2)
7.57×10^{-6}	2.68×10^{-5}

1.2 Nonlinear Friction Model

A more accurate model can be derived by modeling the motor using the Karnop stick-slip model. In this model, the static friction T_c and the breakaway velocity b_v become model parameters. The total friction torque T_f is described by equation 1.15:

$$T_f = \begin{cases} T_{\text{applied}}, & -b_v < \dot{\varphi} < b_v \\ d_m\dot{\varphi} + \text{sign}(\dot{\varphi})T_c, & |\dot{\varphi}| \geq b_v \end{cases} \tag{1.15}$$

By choosing b_v to be an arbitrarily small value, the complexity of the system identification process is reduced. Since the fidelity of the model is still low, the value of b_v is not critical. For this study, $b_v = 10^{-7}$ was deemed sufficient to provide the model with dynamics similar to the motor.

To find appropriate values for the model, a two-step procedure was adopted. First, the model was numerically fitted to the three step responses (5 V, 12 V and 20 V). Second, the model parameters were tuned using a sine wave input. The numerical approach was used solely to accelerate the tuning process.

This approach utilizes a least square method, minimizing Δ in Equation 1.16 with respect to the parameters T_c, d_m, and J_l:

$$\Delta(T_c, d_m, J_l) = (\bar{\tau}_5 - \tau_5)^2 + (\bar{\tau}_{12} - \tau_{12})^2 + (\bar{\tau}_{20} - \tau_{20})^2 + (\bar{K}_5 - K_5)^2 + (\bar{K}_{12} - K_{12})^2 + (\bar{K}_{20} - K_{20})^2. \tag{1.16}$$

Here, the index indicates the amplitude of the step input. The identified parameters result in a model that demonstrated an improved fit to the step responses compared to the nonlinear model, as shown in Figures 4–6. Table 3 illustrates the respective time constant and DC gain with different step inputs.

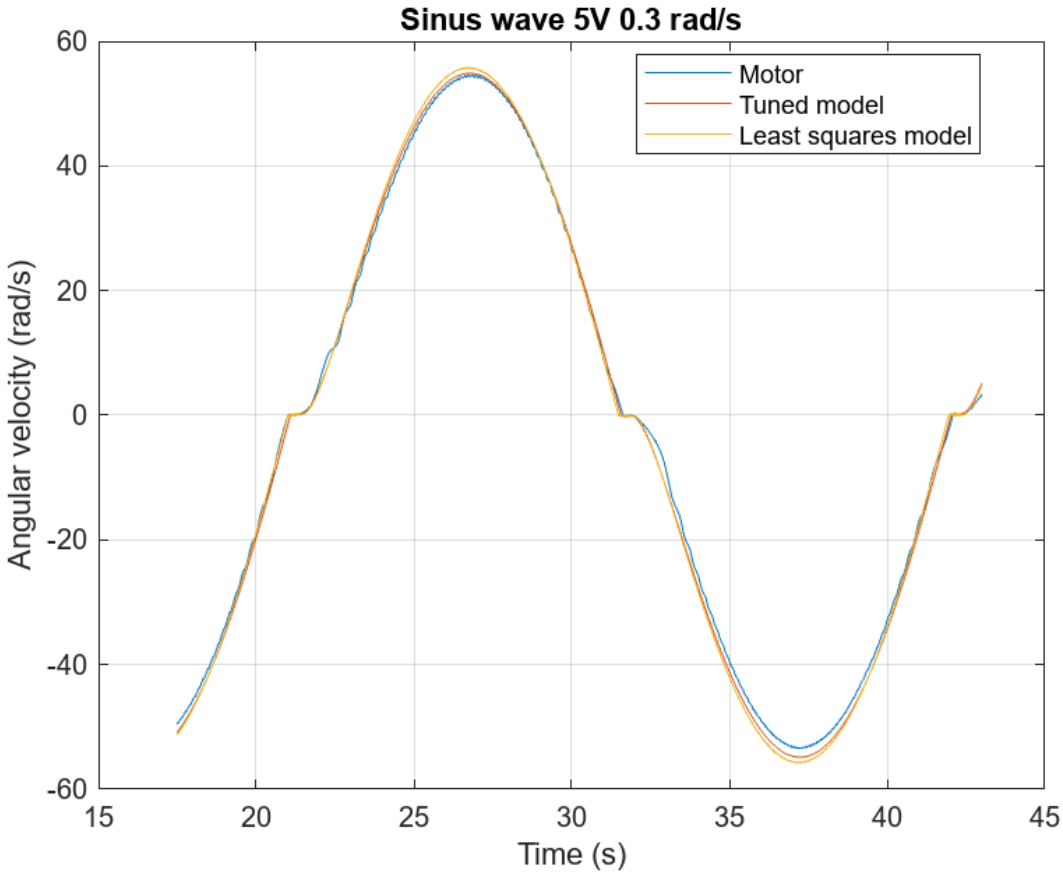

Figure 4: Response comparison for a sinusoidal input with amplitude 5 V and frequency 0.3 rad/s .

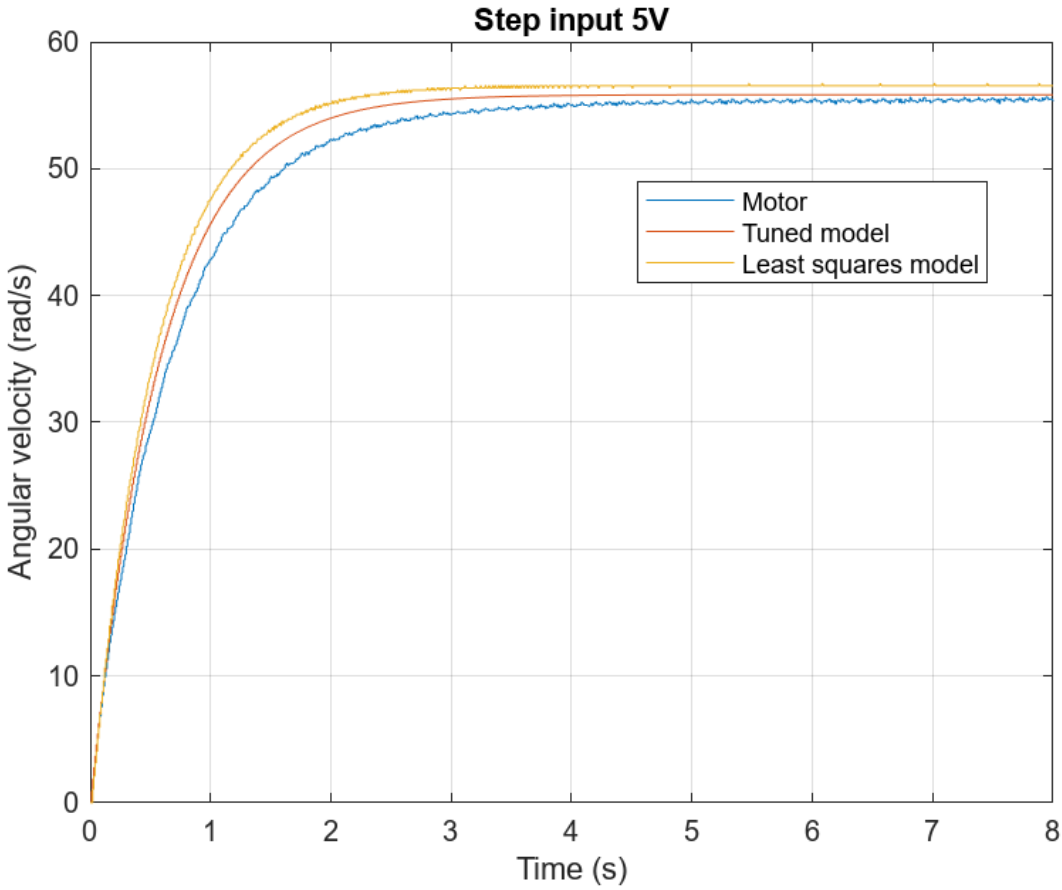

Figure 5: Step response comparison for a 5 V step input .

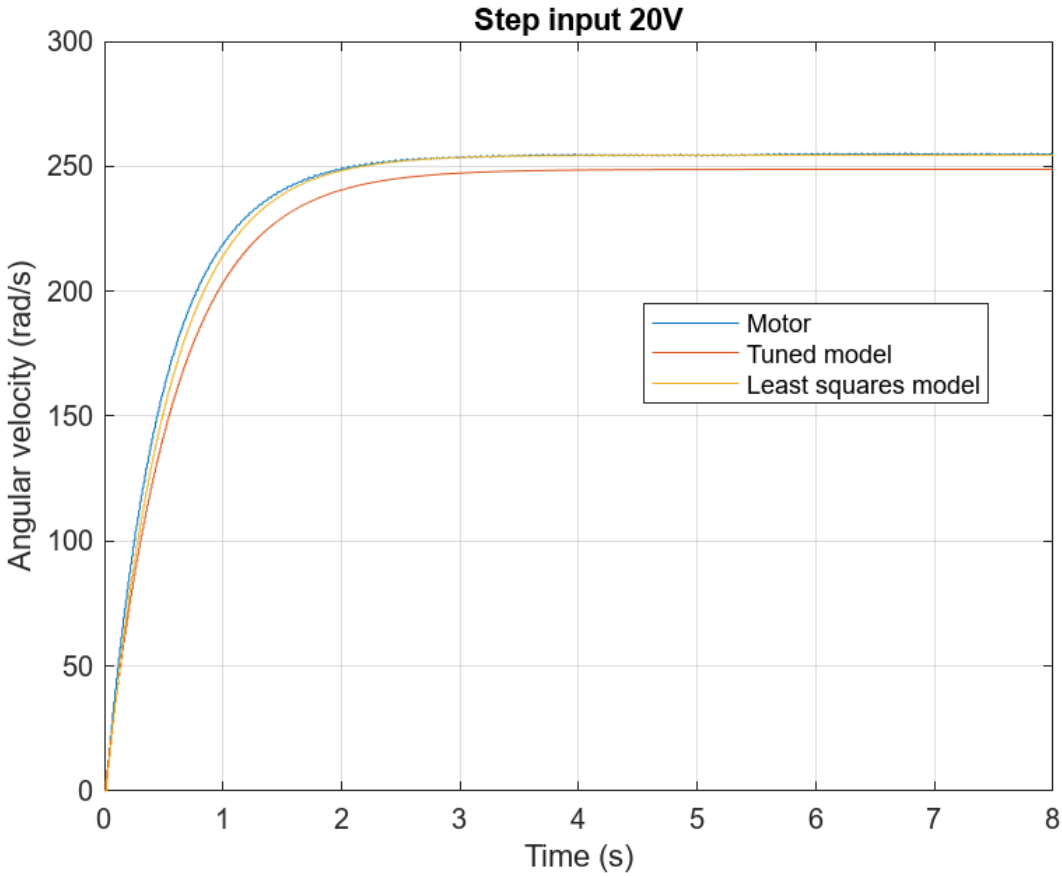

Figure 6: Step response comparison for a 20 V step input.

Table 3: Comparison of DC gains and time constants for motor and nonlinear model.

Input (V)	DC Gain (Motor)	DC Gain (Model)	τ (Motor)	τ (Model)
5	11.3	11.2	0.58	0.58
20	12.8	12.4	0.51	0.58

With the least-squares model, the step responses fit the experimental data well. However, the sinusoidal input case shows less accuracy. To address this, minor tuning were made to the parameters. Specifically, the static friction was slightly reduced, and the damping coefficient was increased. Moreover, adjusting the damping coefficient required modification of the inertia parameter to maintain the overall model dynamics.

Both the least-squares model and the tuned model significantly improve the representation of the motor's behavior compared to the initial linear model. However, discrepancies between the sinusoidal and step input responses were observed. As mentioned in Subsection 1.1.3, the assumption of linear viscous damping does not fully capture the motor's

behavior, which is likely nonlinear with respect to the velocity and influenced by multiple factors such as temperature.

Another interesting observation is the motor's asymmetry in amplitude in the sinusoidal case as shown in Figure 4, suggesting the presence of non-symmetrical damping and static friction. These factors could contribute to some of the observed discrepancies. While a non-symmetrical model was considered to achieve better coverage of the motor's dynamics, it was deemed too resource-intensive to develop.

The current model, as derived, is considered sufficient for designing and verifying a controller with reasonable accuracy. It performs well in the positive direction, and the control performance requirements only focus on positive angular velocity. The resulting parameters are shown in Table 4.

Table 4: Identified parameters of nonlinear model.

d_m (N m s rad^{-1})	J_l (kg m^2)	T_c (N m)
5	2.7803	4.1

Conclusively, due to its implementation of nonlinear characteristics, the model covers a larger range of values that can be reached, simulating the real life tendencies of the motor more closely.

2 Controlling the Motor

2.1 Velocity Control

2.1.1 Formulating Equations

For the velocity controller, the requirements included zero overshoot and maintaining a maximum steady-state error of 1 rad/s for a step input of 50 rad/s. To meet the condition that the controller's transfer function must have the same order in the numerator and denominator, a 2-Degrees of Freedom (DoF) output feedback PI-controller was selected for the ability to reduce the steady-state error to the requirement of 1 rad/s. This is due to the presence of an integrator.

The controller was created using Discrete Digital Design (DDD). This approach meant that the plant was discretized using the triangle (First Order Hold (FOH)) approximation, enabling a larger sample time to be utilized compared to the Design by Emulation (DBE) approach. By substituting in the values from Table 2 into Equation 1.8, the transfer function in continuous time became

$$\frac{\Omega(s)}{U(s)} = \frac{22.58}{s + 1.849} \tag{2.1}$$

and in discrete (z) time using the triangle (FOH) approximation.

$$\frac{\Omega(z)}{U(z)} = \frac{0.6603}{z - 0.9459} \tag{2.2}$$

The latter was then described using notations as in the Diophantine equation

$$\frac{B}{A} = \frac{0.6603}{z - 0.9459} \tag{2.3}$$

The PI-controller was then constructed in discrete time as well, also utilizing the Diophantine notations. First, the feedback part of the controller was implemented:

$$\frac{S}{R} = \frac{K_P z + K_I}{z - 1} \tag{2.4}$$

Second, the feedforward part:

$$\frac{T}{R} = \frac{t_0 A_o}{z - 1} \tag{2.5}$$

where

$$A_o = z - e^{\omega_o T_s} \tag{2.6}$$

and ω_o is the pole of the observer in continuous time and T_s is the sampling frequency, often chosen to be approximately 10-30 times faster than the fastest pole in the system (feedback part, feedforward part, plant or closed loop system). By taking the limit

$$\lim_{z \to 1} \frac{TB}{AR + BS} = 1, \tag{2.7}$$

an additional condition is created where the DC-gain equals 1. Together with the Diophantine equation:

$$AR + BS = A_m A_o, \tag{2.8}$$

where

$$A_m = z - e^{\omega_m T_s} \tag{2.9}$$

and ω_m is the desired pole of the system in continuous time (A_m needs to be the same order as the denominator of the plant, A), a system of equations can be created with three unknowns, i.e. K_P, K_I and t_0, and three equations.

To further reduce oscillation and mitigate overshoot, anti-windup was incorporated into the controller, which is shown in the block diagram below.

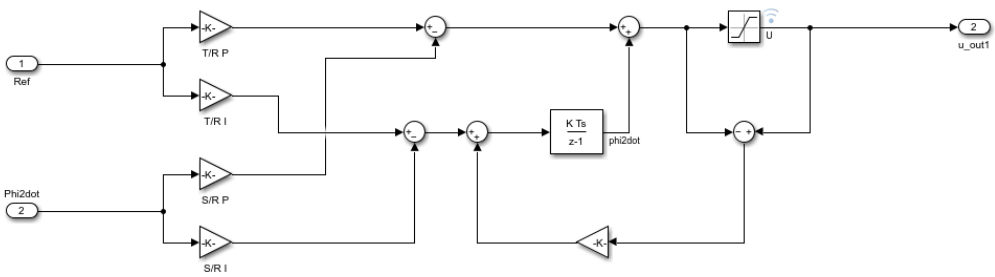

Figure 7: Block diagram of PI controller and anti-windup.

This adjustment ensures that the integral accumulation from both the feedback and forward parts were adjusted when the controller output reached saturation limits, preventing the integrator from excessive error accumulation.

By evaluating the performance of the system using the DC motor model derived previously, the following poles (in continuous time) were chosen together with a sampling frequency (chosen to be 19 times larger than the fastest pole in the system) to match the performance requirements previously stated. These values are shown in the table below.

Table 5: Values of placed poles (in continuous time) and sample period.

ω_m	ω_o	T_s
-11 rad/s	-4 rad/s	0.0301 sec

Solving the system of equations (Equation 2.7 and Equation 2.8), the values of K_P, K_I and t_0 could then be derived in discrete time. These are shown in the following table.

Table 6: Values of PI constants and t_0 (in discrete time).

K_P	K_I	t_0
0.537	-0.487	0.441

The following figure shows the Pole-Zero map of the closed-loop system in continuous time.

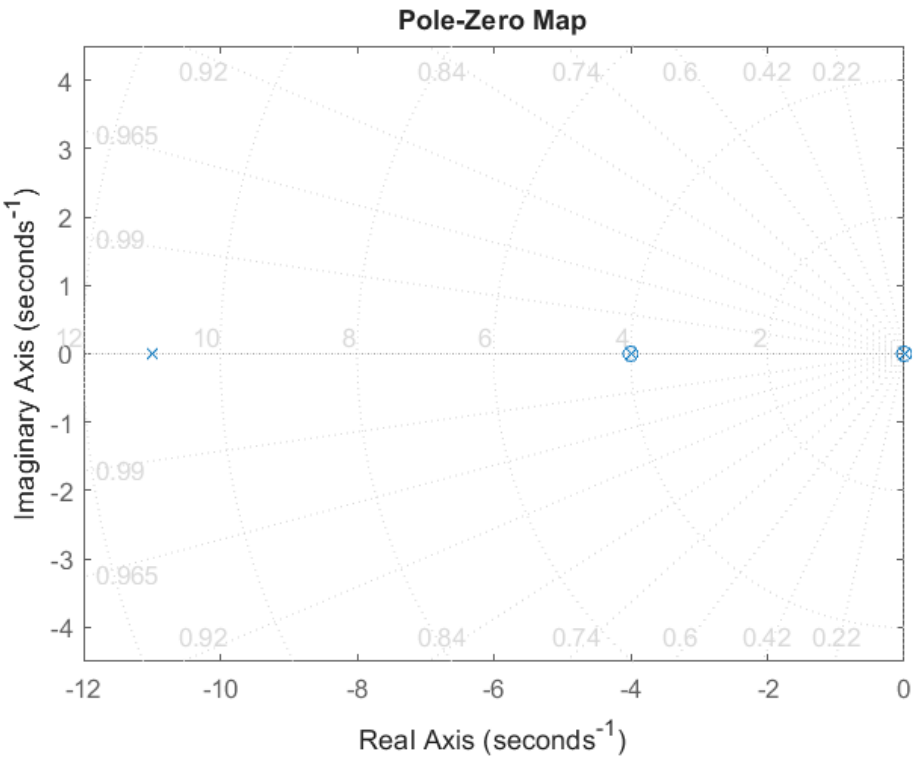

Figure 8: Pole-Zero map for the closed loop system.

The visualization helps to better understand the nature and the placements of the poles. The poles are correctly positioned as specified and match the values listed in Table 5. The resulting transfer function of the output feedback controller was the following.

$$u(z) = \frac{0.441(z - 0.887)}{z - 1} ref(z) - \frac{0.537z - 0.487}{z - 1} vel(z) \tag{2.10}$$

where the A_o pole was converted from continuous time (-4 rad/s) to discrete time (0.887).

2.1.2 Performance Validation on Physical System

The velocity controller was tested for three different cases.

Case 1 First a step input of 50 rad/s. The following figure shows the model velocity, and motor measurement velocity.

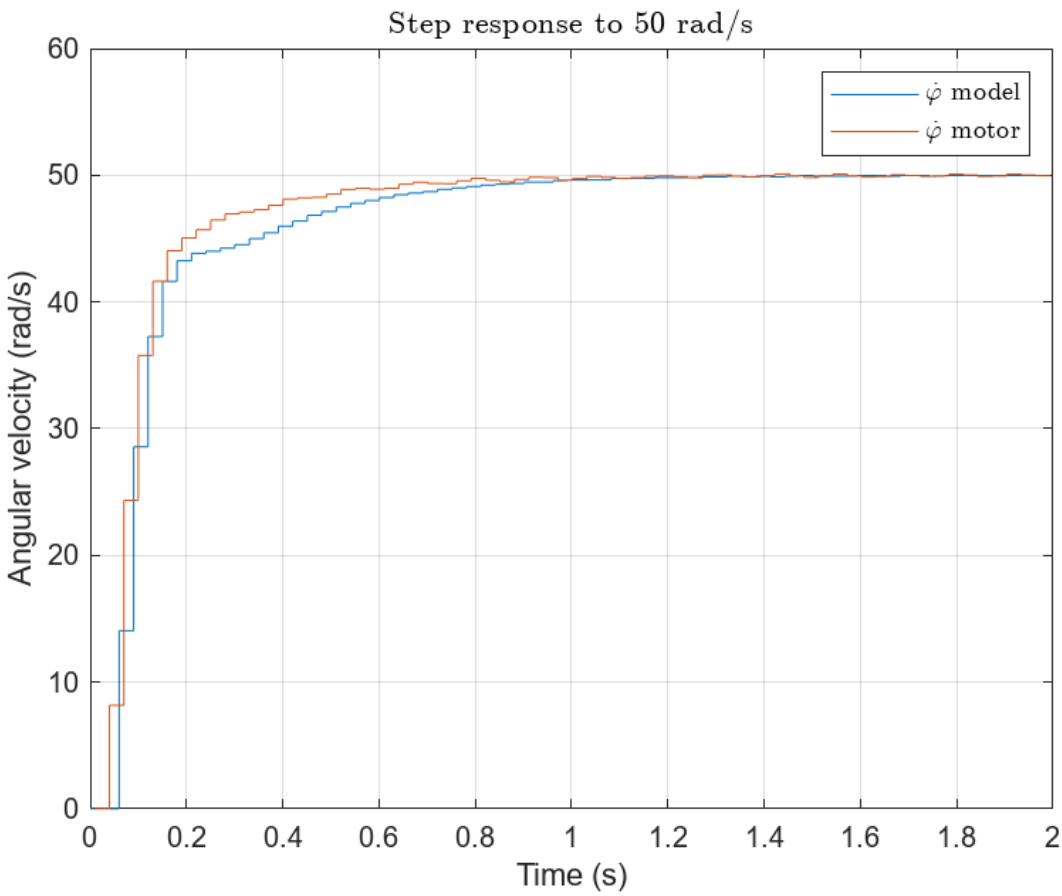

Figure 9: Model velocity and motor measurement velocity for a step input of 50 rad/s.

From Figure 9, it can be observed that there is no overshoot and the rise time is approximately 0.16 seconds. In the following figure, it shows that the steady-state error is less than 1 rad/s, and the settling time is approximately 0.59 seconds.

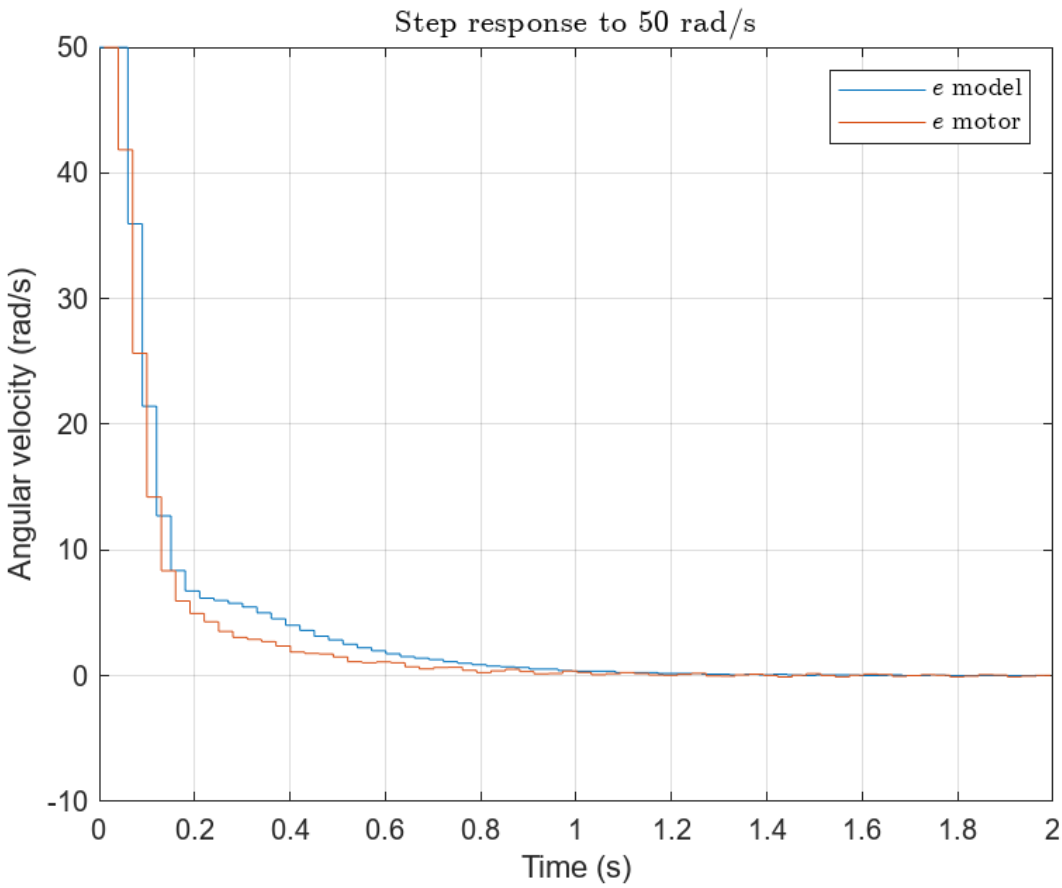

Figure 10: Motor velocity error and model velocity error for a step input of 50 rad/s.

Furthermore, a plot of the voltage is shown in Figure 11.

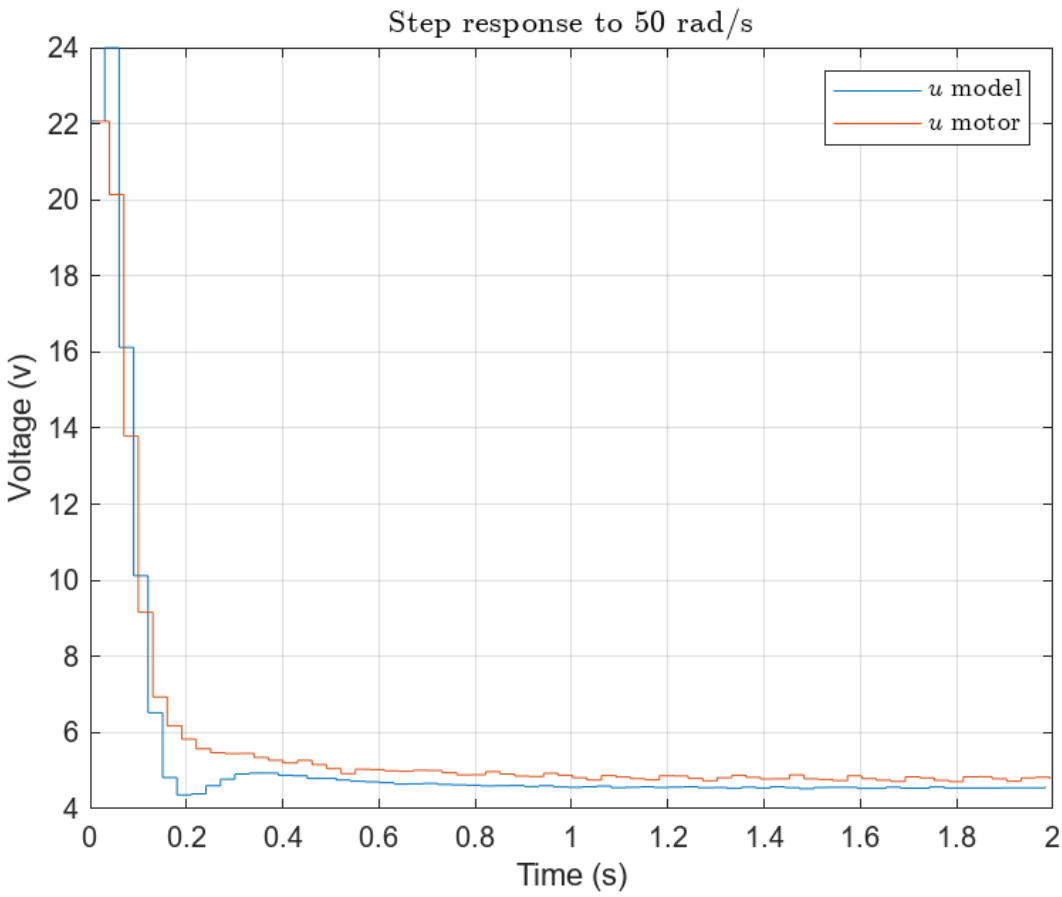

Figure 11: Motor and model supply voltage for a step input of 50 rad/s.

For the model, the voltage initially saturates at 24 V, before settling at 4.5 V to maintain the steady-state velocity. However, for the real motor, the voltage never saturates, reaching a maximum of 22 V before also settling at around 4.5 V.

Case 2 Next, the controller was tested with a sinusoidal reference signal with an amplitude of 50 rad/s and a frequency of 0.2 Hz. Figure 12 illustrates the results, showing the speed reference signal, model velocity and the motor measurement velocity. The model closely follows the reference signal and the motor velocity aligns well with the model's behavior.

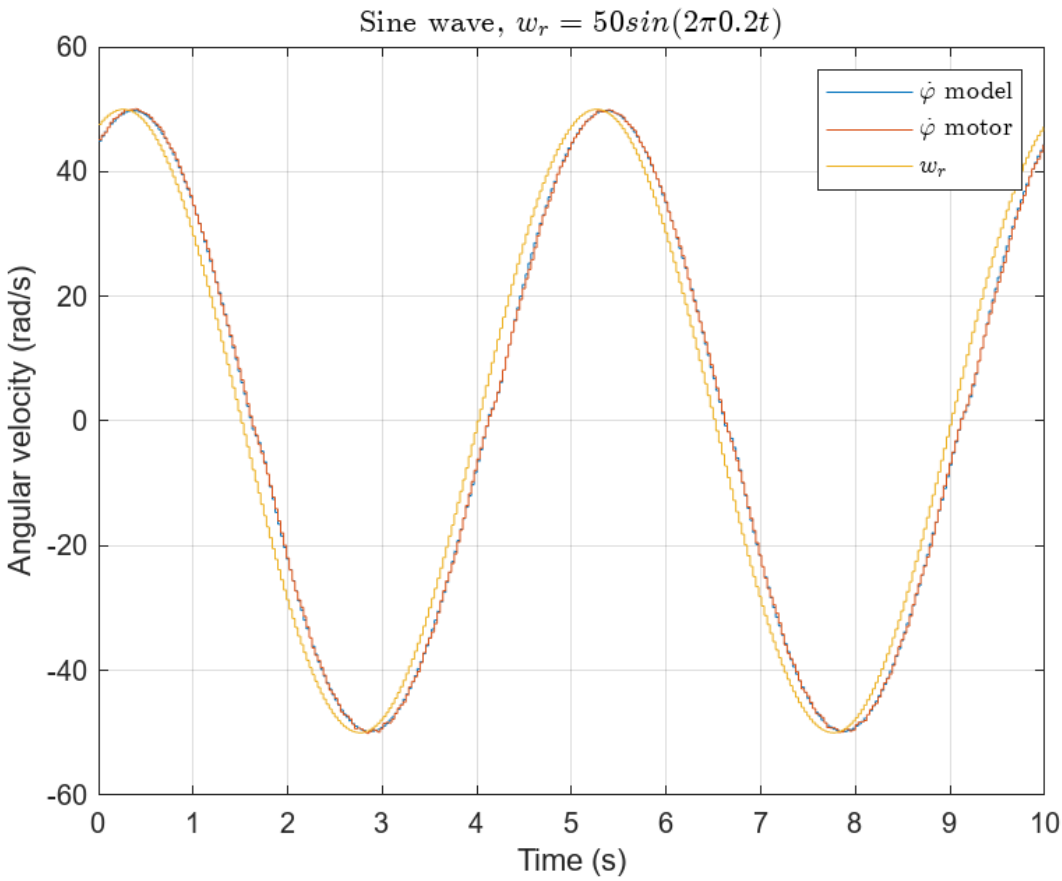

Figure 12: Speed reference signal, model velocity and motor measurement velocity for a sine input with 50 rad/s amplitude and 0.2 Hz frequency.

Shown in the next figure, the error plot reveals that the error oscillates with a maximum absolute value of approximately 9 rad/s for both the model and the motor, occurring near the inflection points of the sinusoidal input.

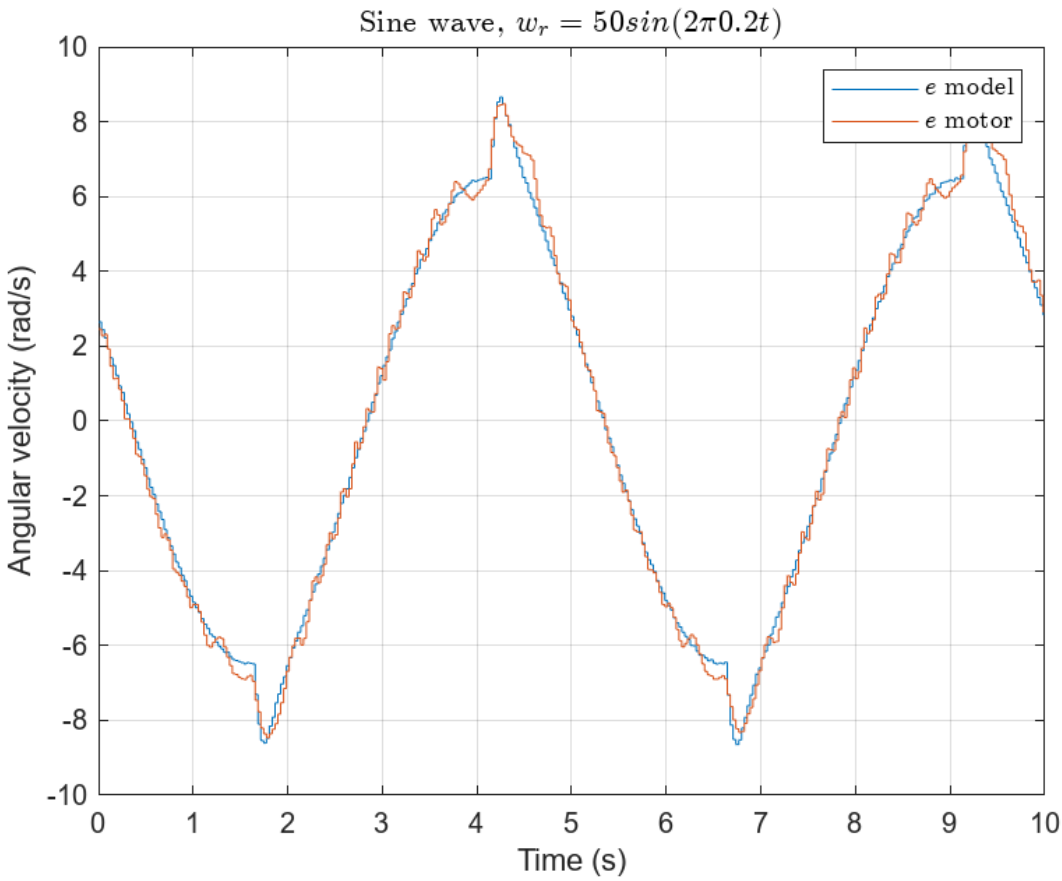

Figure 13: Motor velocity error for a sine input with 50 rad/s amplitude and 0.2 Hz frequency.

Figure 14 illustrates the motor supply voltage, which roughly follows the profile of a sine wave, oscillating between 6 V and -6 V for the real motor, and 5.5 V and -5.5 V for the model.

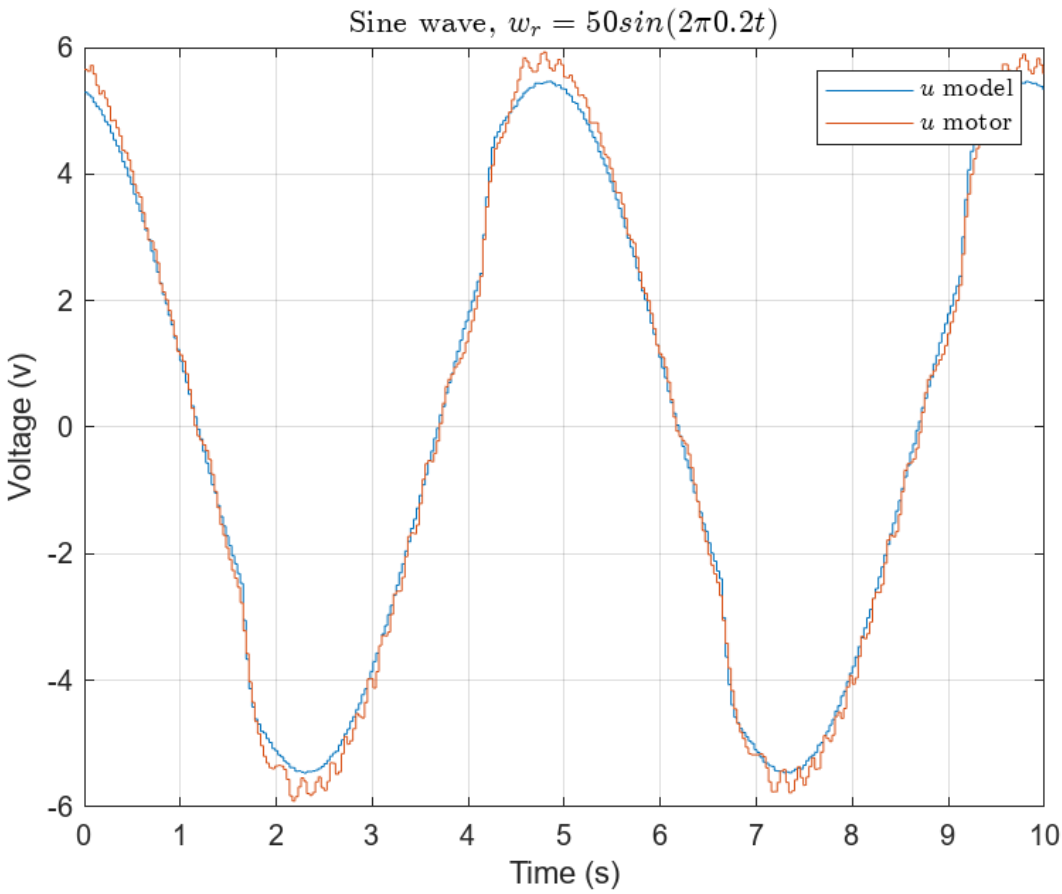

Figure 14: Motor and model supply voltage for a sine input with 50 rad/s amplitude and 0.2 Hz frequency.

Case 3 To analyze the system's behavior under disturbances, the same reference signal was applied, but disturbances were introduced into the system. Figure 15 illustrates the motor measurement velocity in response to the disturbances, showing deviations from the reference signal while force is applied on the flywheel.

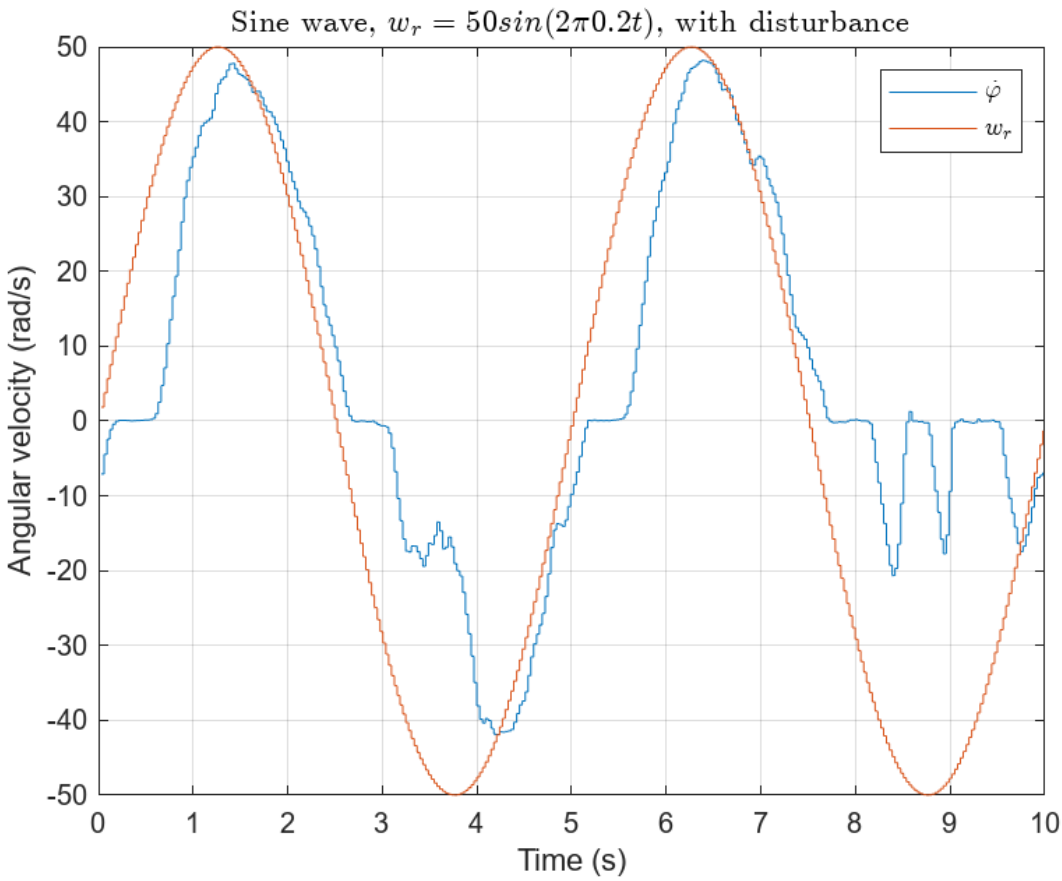

Figure 15: Speed reference signal and motor measurement velocity for a sine input with 50 rad/s amplitude and 0.2 Hz frequency when exposed to disturbances.

In Figure 16, the error is shown.

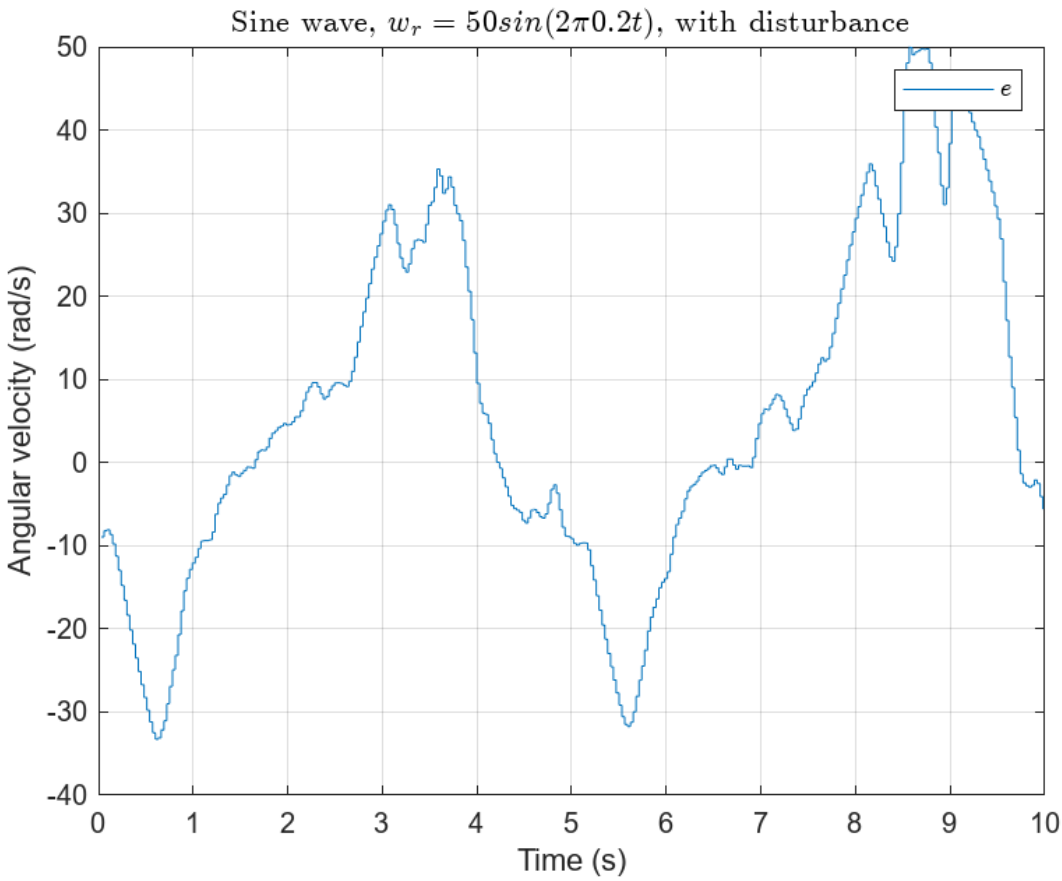

Figure 16: Motor velocity error for a sine input with 50 rad/s amplitude and 0.2 Hz frequency when exposed to disturbances.

Figure 17 shows the motor supply voltage with the presence of disturbance. The disturbances caused the voltage to saturate at certain points.

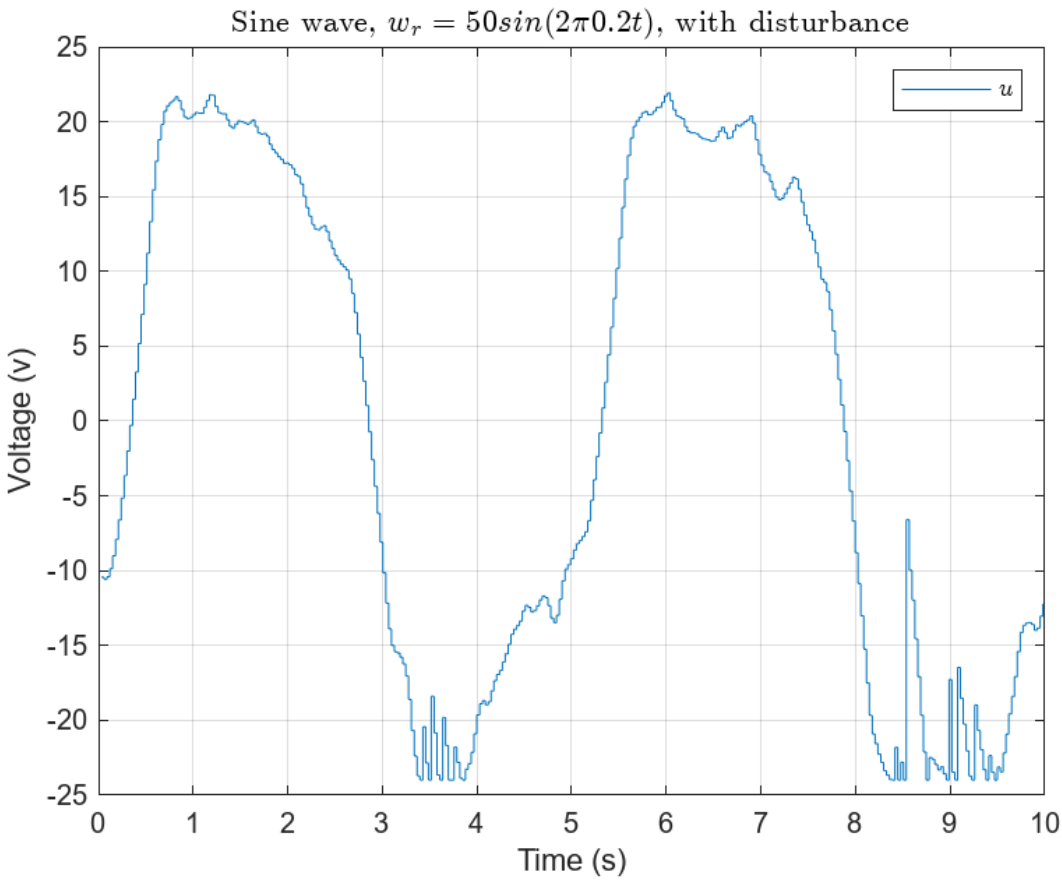

Figure 17: Motor supply voltage for a sine input with 50 rad/s amplitude and 0.2 Hz frequency when exposed to disturbances.

When analyzing Figure 16 and Figure 17, it seems saturation occurs when the error reaches an absolute value of approximately 20 rad/s.

2.2 Position Control

2.2.1 Formulating Equations

For the position controller, the requirements include a step response from 0 to 10 rad with a rise time of 0.5 s ± 0.05 s, where rise time is defined as time between 10% and 90% of the reference step; a maximum overshoot of 2% of the step amplitude; and a maximum 0.5% steady-state error. Additionally, to fulfill the requirements from level 2, the sample period needs to be as long as possible, and the controller needs to be designed in discrete time via a DDD approach.

To meet the requirements, a 2-DoF output feedback PID-controller was selected. Compared to a PI-controller, it adds a derivative part, which improves the system's ability to

predict and counteract rapid changes in error, enhancing stability and reducing overshoot and oscillations. With a derivative term, a low-pass filter part needs to be added as well to attenuate high-frequency noise that can be amplified due to the derivative, ensuring smooth and robust control.

Using a DDD approach, again, meant that the plant and controller were discretized from the beginning, enabling a larger sample time to be utilized compared to the DBE approach. Substituting the values from Table 2 into Equation 1.5 and simplifying, the transfer function in continuous time became:

$$\frac{\Phi(s)}{U(s)} = \frac{21.8}{s^2 + 1.785s} \tag{2.11}$$

and in discrete (z) time:

$$\frac{\Phi(z)}{U(z)} = \frac{0.0008264z + 0.0008221}{z^2 - 1.985z + 0.9845}. \tag{2.12}$$

The latter was then described using notations as in the Diophantine equation

$$\frac{B}{A} = \frac{0.6603}{z - 0.9459} \tag{2.13}$$

The PID-controller was constructed in discrete time as well, utilizing the Diophantine notations. First, the feedback part of the controller was implemented:

$$\frac{S}{R} = \frac{K_D z^2 + K_P z + K_I}{(z-1)(z+L)} \tag{2.14}$$

where L represents the cutoff frequency of the low-pass filter. Second, the feedforward part:

$$\frac{T}{R} = \frac{t_0 A_0}{z - 1} \tag{2.15}$$

where

$$A_o = (z - e^{-b_+ T_s})(z - e^{-b_- T_s}) \tag{2.16}$$

and

$$b_{+/-} = \zeta_o \omega_o \pm i\omega_o \sqrt{1 - \zeta_o^2} \tag{2.17}$$

ζ_o is the damping coefficient of the observer poles.

By taking the limit

$$\lim_{z \to 1} \frac{TB}{AR + BS} = 1, \tag{2.18}$$

an additional condition is created where the DC-gain equals 1. Together with the Diophantine equation:

$$AR + BS = A_m A_o, \tag{2.19}$$

where

$$A_m = (z - e^{-a_+ T_s})(z - e^{-a_- T_s}) \tag{2.20}$$

and

$$a_{+/-} = \zeta_m \omega_m \pm i\omega_m \sqrt{1 - \zeta_m^2}, \tag{2.21}$$

ω_m is the desired pole of the system in continuous time (A_m needs to be the same order as the denominator of the plant, A) and ζ_m is the damping coefficient of the system; a system of equations can be created with five unknowns, i.e. K_P, K_I, K_D, L and t_0, and five equations.

By evaluating the derived motor model, the following poles (in continuous time) were chosen together with an as-long-as-possible sample period (chosen to be 2.65 times larger than the fastest pole in the system). Using a larger sample period leads to too slow of a rise time, however, with an overshoot that is still under 2%. These values are shown in the table below.

ω_m	ω_o	ζ_m	ζ_o	T_s
-6 rad/s	-24 rad/s	0.9	1	0.0988 sec

Table 7: Values of placed poles, damping coefficients and sample period (in continuous time).

Solving the system of equations (Equation 2.18 and Equation 2.19), the values of K_P, K_I, K_D, L and t_0 could then be derived. These are shown in the following table.

K_P	K_I	K_D	L	t_0
-13.63	5.126	9.386	0.575	1.076

Table 8: Values of PID constants, low-pass filter constant and t_0 (in discrete time).

To further reduce oscillations and overshoot, the output of the integrator was constrained. Specifically, the integral for both the feedback and forward components was limited to a certain saturation limit, ±50. This is shown in the following block diagram.

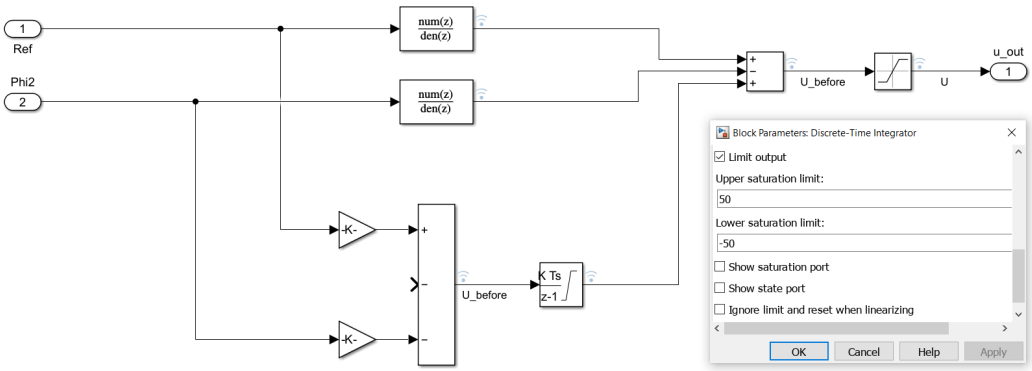

Figure 18: Block diagram of PID controller, low-pass filter and integral saturation limit.

The resulting transfer function of the output feedback controller was the following.

$$u(z) = \frac{1.076(z - 0.0934)}{(z-1)(z+0.575)} ref(z) - \frac{9.386z^2 - 13.63z + 5.126}{(z-1)(z+0.575)} pos(z) \tag{2.22}$$

where the A_o pole was converted from continuous time (-24 rad/s) to discrete time (0.0934).

2.2.2 Performance Validation on Physical System

As shown in Figure 19, the position controller was tested with a step input reference of 10 rad.

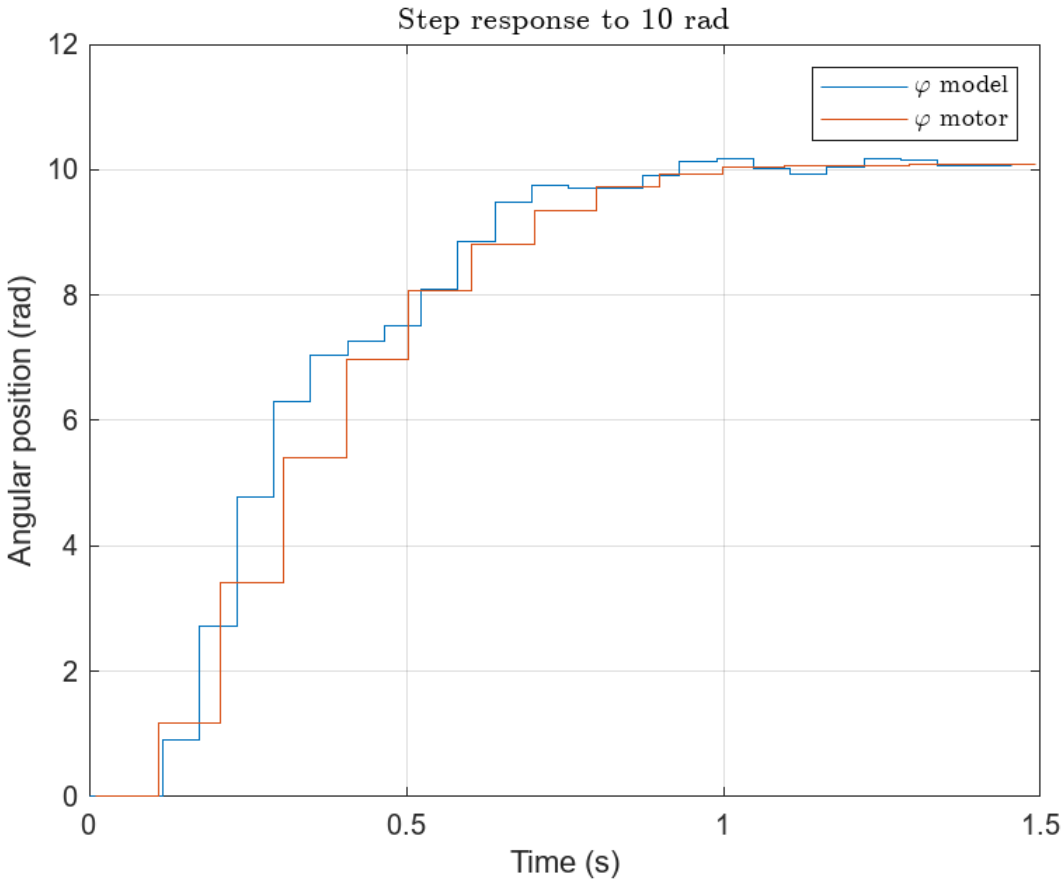

Figure 19: Model position and motor measurement position for a step input of 10 rad/s.

Seen in the figure, the largest overshoot was produced by the model and was approximately 0.77%, where the rise time was approximately 0.54 seconds which follows the requirements. As shown in Figure 20, the steady-state error reaches a value much less than 0.5% for both the real motor and the model, where the settling time is approximately 0.83 seconds.

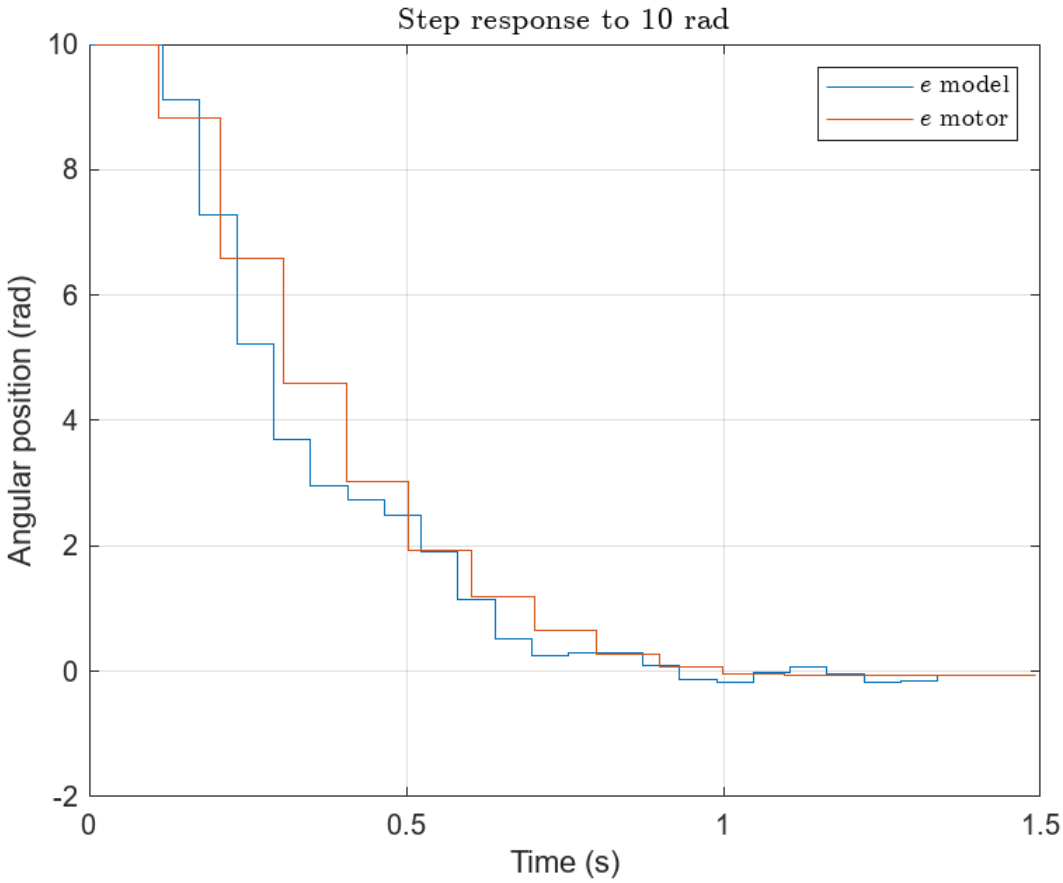

Figure 20: Model position error and motor measurement position error for a step input of 10 rad/s.

The corresponding motor supply voltage is shown in Figure 21. The voltage never saturates, only reaching a maximum value of ca 14 V for the model, and 11 V for the motor.

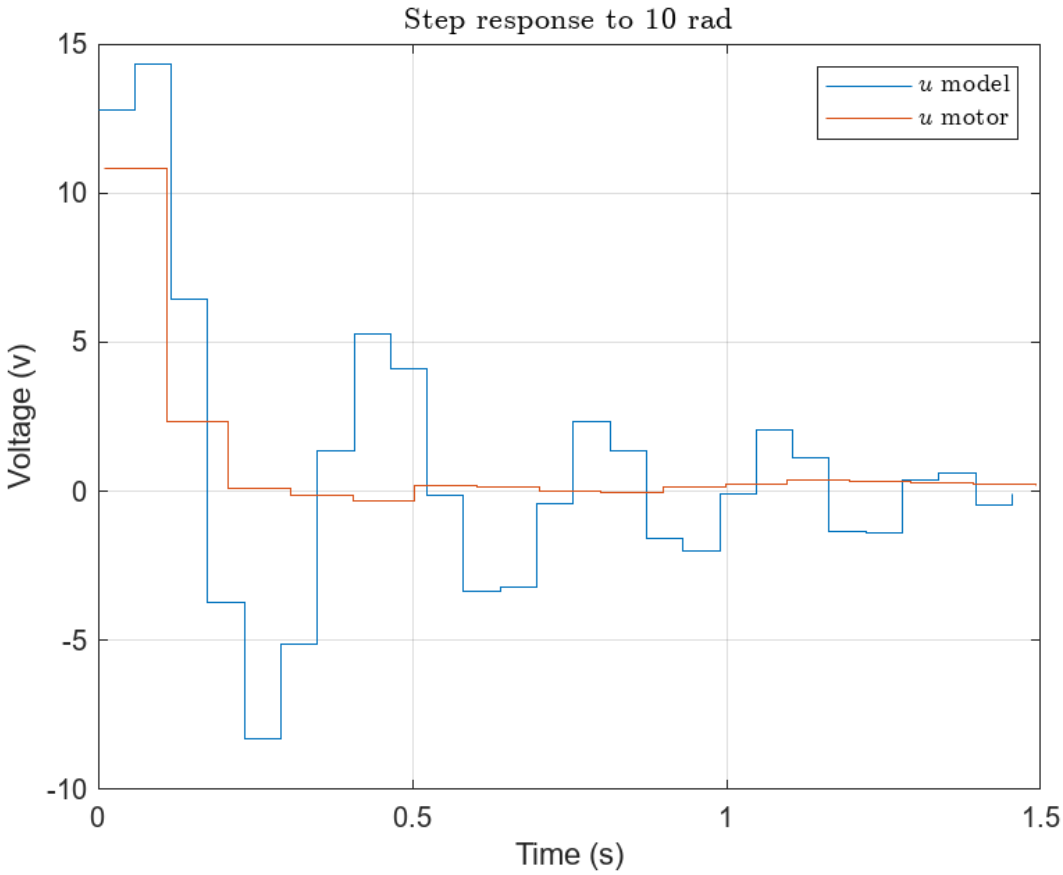

Figure 21: Motor and model supply voltage for a step input of 10 rad/s.

2.2.3 Discussion

The sampling period of a controller depends on several factors. First, it must follow the Nyquist-Shannon theorem, meaning it should be at least twice as fast as the highest frequency in the system to avoid errors like aliasing. In this case, the fastest pole is ω_o, and the slowest sampling frequency able to meet the performance requirements was deemed to be just above twice as fast, 2.65. It is possible that a slightly slower observer pole was possible to implement and thereby a larger sampling period. However, what in the end limits the lowest deployable sampling frequency is the performance requirements.

Faster systems or controllers require shorter sampling periods to work accurately and respond quickly. However, practical issues like delays in computation, sensors, actuators, and hardware limitations can also influence the choice of sampling period. Very short sampling periods can make a system more prone to oversampling, especially in systems using derivatives. Finding the right balance is key to keeping the system effective and stable.

Using a discrete digital design (DDD) approach for the controller allows for longer sampling periods compared to design by emulation (DBE). This is because DDD directly

designs the controller for the discrete, sampled system rather than converting a continuous design. This approach reduces performance issues and instability that can come from sampling effects or approximations. As a result, the DDD method provides more flexibility when choosing the sampling period while still ensuring reliable and accurate control.